

Project Report: Real-Time Pothole Detection Using YOLOv8

1. Executive Summary

This project successfully develops and deploys a custom computer vision model capable of detecting road potholes in real-time. By utilizing the state-of-the-art **YOLOv8s (Small)** architecture and training it on a custom dataset, the system achieves a balance between computational efficiency and high detection accuracy. The final model is fully optimized for practical deployment, converting raw video feeds into actionable data by automatically mapping bounding boxes around road hazards.

2. System & Hardware Architecture

To ensure high-performance training and real-time inference, the model was trained using modern hardware acceleration.

- **Graphics Processing Unit (GPU):** NVIDIA GeForce RTX 4050
- **Deep Learning Framework:** PyTorch (CUDA Accelerated)
- **Development Environment:** Visual Studio Code (VS Code)
- **Operating System Platform:** Windows 11

3. Dataset Methodology & Configuration

The AI model was trained using a structured dataset containing specialized annotations for road anomalies.

Dataset Distribution

- **Total Images:** 1,977 images
- **Training Set (80%):** 1,581 images (*used by the AI to learn core visual patterns*)
- **Validation Set (20%):** 396 images (*used to independently evaluate performance at each epoch*)

Configuration File (data.yaml)

YAML

train: ./train/images

val: ./valid/images

nc: 1

names: ['pothole']

4. Model Training Hyperparameters

The training script was built using Python to dynamically establish absolute file system paths, mitigating standard Windows-specific path resolution issues.

- **Model Architecture:** yolov8s.pt (YOLOv8 Small - optimized for edge deployment)
- **Training Execution:** 50 Complete Epochs
- **Image Resolution Size:** 640 × 640 pixels
- **Batch Size:** 16
- **Worker Threads:** 2 (*optimized to eliminate Windows multi-threading data bottlenecks*)
- **Memory Optimization:** Enabled (cache=True to store images directly in RAM for faster epoch transitions)

5. Performance Evaluation & Results

Confusion Matrix Analysis

Upon reaching the 50th epoch, the training metrics stabilized, and the final model weights (best.pt) were saved. The performance was validated using a normalized confusion matrix.

- **True Positive Rate (Pothole → Pothole): 0.79 (79%)**
- **Analysis:** When a real-world pothole is present on the road, the custom model successfully detects and identifies it 79% of the time. This indicates strong feature extraction capabilities, allowing the AI to effectively differentiate between standard asphalt textures, shadows, and actual road damage.

6. Real-Time Inference & Deployment

The finalized model was deployed via an execution script (predict.py) to process dynamic video data (sample_video.mp4).

The system smoothly runs inference on the input video, rendering bounding boxes overlaid with confidence percentages directly onto the frames. The final output is automatically compiled and saved as an accelerated video file inside the project's directory:

```
\runs\detect\predict\sample_video.avi
```

7. Conclusion & Future Scope

The project demonstrates that a localized object detection model can be successfully trained to an accuracy level of nearly 80% using accessible consumer hardware like the RTX 4050.

Future Enhancements:

1. Expanding the dataset to include varying weather conditions (rainy/wet asphalt, night driving).
2. Deploying the weight file (best.pt) to a low-power edge device (like a Raspberry Pi or Jetson Nano) paired with a dashcam for real-time in-car alerting systems.

